POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 1002 | Semester I | Theory | 4 Credits

Fundamentals of Engineering Mathematics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Diploma in Engineering and Technology**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week; 60 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **1002**, titled **Fundamentals of Engineering Mathematics**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

> **Source treatment:** PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

> **Verified source note:** No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

This foundation course develops mathematical tools used in engineering: matrices, trigonometry, coordinate geometry and differential calculus. The emphasis is on accurate symbolic work, clear diagrams and a reproducible problem-solving method.

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Prerequisite foundation

Linear equations, quadratic equations, rectangular coordinates and angles from Secondary Education.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week; 60 h total	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Apply matrices and determinants to linear equations.
2. Use trigonometric concepts and relationships.
3. Represent and interpret straight lines and circles.
4. Evaluate limits and derivatives.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Apply matrices and determinants to solve linear systems with two and three unknowns.	Apply
CO2	Use trigonometric identities and functions to solve trigonometric problems.	Apply
CO3	Interpret straight-line and circle problems using coordinate geometry.	Apply
CO4	Evaluate limits and derivatives.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Matrices and Determinants	Matrix concept and types; transpose; algebra; determinants; minors/cofactors; adjoint/inverse of 2×2 matrix; Cramer's rule.	11 L + 4 T hours	CO1	Module 1
2	Trigonometry	Angles; ratios; standard angles; Pythagorean relations; ASTC and reduction; compound angles.	11 L + 4 T hours	CO2	Module 2
3	Coordinate Geometry	Slope and slope-point line; angle between lines; parallel/perpendicular lines; circle equation, centre and radius.	11 L + 4 T hours	CO3	Module 3
4	Differential Calculus	Limits by substitution/factorisation; special forms; derivative concept and standard results; rules and second derivatives.	12 L + 3 T hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Matrices and Determinants

Matrix concept and types; transpose; algebra; determinants; minors/cofactors; adjoint/inverse of 2×2 matrix; Cramer's rule.

CO1 · 11 L + 4 T hours

Trigonometry

Angles; ratios; standard angles; Pythagorean relations; ASTC and reduction; compound angles.

CO2 · 11 L + 4 T hours

Coordinate Geometry

Slope and slope-point line; angle between lines; parallel/perpendicular lines; circle equation, centre and radius.

CO3 · 11 L + 4 T hours

Differential Calculus

Limits by substitution/factorisation; special forms; derivative concept and standard results; rules and second derivatives.

CO4 · 12 L + 3 T hours

MODULE 1

Matrices and Determinants

Matrix concept and types; transpose; algebra; determinants; minors/cofactors; adjoint/inverse of 2×2 matrix; Cramer’s rule.

11 L + 4 T hours **CO1** **Understand · Apply**

Learning goals

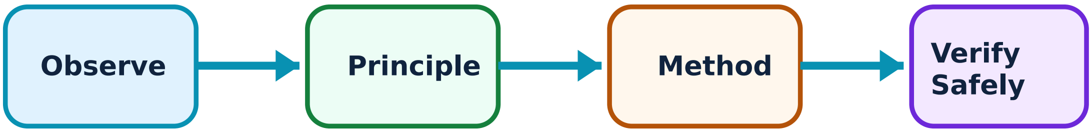
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Matrices and Determinants: identify the system, state its principle, follow the correct method and verify the result safely.

1. Matrices, order and special types−

A matrix is a rectangular arrangement of entries. Its order is rows by columns. The official types include row, column, square, diagonal, triangular, identity and zero matrices. Type and order determine which operations are meaningful.

Study and application method

Write order first, identify the requested type by checking positions of entries, and preserve brackets when presenting a matrix.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write order first, identify the requested type by checking positions of entries, and preserve brackets when presenting a matrix.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Matrix addition/subtraction requires equal order; do not add entries from incompatible matrices.

2. Transpose and algebra of matrices−

Transpose interchanges rows and columns. Matrix algebra includes scalar multiplication, equality, addition, subtraction and multiplication. Matrix multiplication is an ordered row-by-column operation and is generally not commutative.

Study and application method

For multiplication, check inner dimensions, compute each entry as row-dot-column and state final order.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For multiplication, check inner dimensions, compute each entry as row-dot-column and state final order.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: [illegible] concept [illegible] [illegible] [illegible]. [illegible] diagram / circuit / formula / procedure [illegible] [illegible]. [illegible] result [illegible] application [illegible] [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Common mistake / precaution: AB need not equal BA; never reverse order without checking dimensions and result.

3. Determinants, inverse and Cramer's rule−

A determinant is associated with a square matrix. Minors and cofactors support adjoint construction; a 2×2 inverse exists when determinant is nonzero. Cramer's rule solves a compatible linear system through determinant ratios.

Study and application method

Define coefficient determinant, calculate systematically, check nonzero condition and substitute each determinant in correct variable ratio.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

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Common mistake / precaution: An inverse cannot be used when determinant is zero; write this condition explicitly.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Trigonometry

Angles; ratios; standard angles; Pythagorean relations; ASTC and reduction; compound angles.

11 L + 4 T hours | **CO2** | **Understand · Apply**

Learning goals

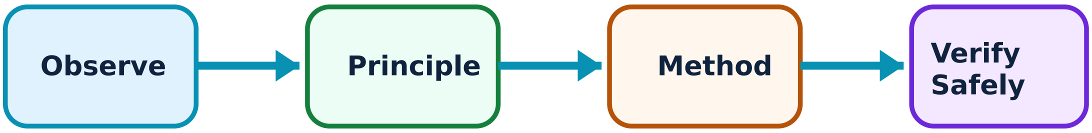
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Trigonometry: identify the system, state its principle, follow the correct method and verify the result safely.

1. Angles and trigonometric ratios−

Angles may be acute, obtuse, right, straight, reflex, complete, positive or negative. Sine, cosine and tangent ratios relate sides of an acute right triangle, while reciprocal ratios complete the standard set.

Study and application method

Sketch the angle, state quadrant/sign and use a known ratio with Pythagorean relation when another ratio is needed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the angle, state quadrant/sign and use a known ratio with Pythagorean relation when another ratio is needed.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Keep angle unit and quadrant convention clear; a reference-angle value may need a sign change.

2. ASTC, reduction formulae and identities−

Pythagorean relations and ASTC rule organise signs and relationships of trigonometric functions in all quadrants. Reduction formulae convert related angles to a standard reference.

Study and application method

Reduce angle to a familiar form, establish sign from quadrant and then apply identity with each algebraic step shown.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Reduce angle to a familiar form, establish sign from quadrant and then apply identity with each algebraic step shown.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not copy only numerical standard-angle value before deciding the sign.

3. Compound angles−

Sine, cosine and tangent compound-angle formulae relate functions of A±B to functions of A and B. They support exact evaluation and engineering geometry calculations within official scope.

Study and application method

Write the selected formula first, substitute exact values and simplify in a controlled order.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write the selected formula first, substitute exact values and simplify in a controlled order.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: The sign inside a compound-angle formula is not always the same as the sign outside a term; write brackets.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Coordinate Geometry

Slope and slope-point line; angle between lines; parallel/perpendicular lines; circle equation, centre and radius.

11 L + 4 T hours **CO3** **Understand · Apply**

Learning goals

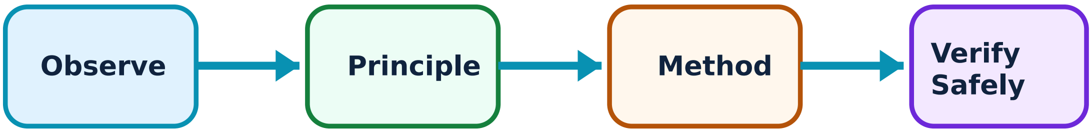
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Coordinate Geometry: identify the system, state its principle, follow the correct method and verify the result safely.

1. Slope and equation of a straight line−

Slope measures line inclination. The slope-point form creates a line equation from one point and a known slope. It connects algebraic equation to a plotted geometric object.

Study and application method

Find slope from points or angle, substitute a chosen point into point-slope form and simplify only after expanding carefully.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Find slope from points or angle, substitute a chosen point into point-slope form and simplify only after expanding carefully.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Vertical lines have undefined slope; do not force them into an ordinary finite-slope calculation.

2. Angle, parallel and perpendicular lines−

The angle between lines depends on their slopes. Parallel lines have equal slopes; perpendicular non-vertical lines have slopes whose product is −1.

Study and application method

Extract slopes from standard equation, select correct relation and check result against sketch.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Extract slopes from standard equation, select correct relation and check result against sketch.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A negative reciprocal rule needs care if either line is vertical or horizontal.

3. Circles and their general equation−

A circle is the locus of points at constant distance from centre. The general equation $x^2+y^2+2gx+2fy+c=0$ yields centre $(-g,-f)$ and radius from its coefficients when valid.

Study and application method

Compare coefficients term by term, calculate centre and radius, then verify by completing square or substitution where appropriate.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare coefficients term by term, calculate centre and radius, then verify by completing square or substitution where appropriate.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A negative radius-squared result has no real circle; report the interpretation.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Differential Calculus

Limits by substitution/factorisation; special forms; derivative concept and standard results; rules and second derivatives.

12 L + 3 T hours | CO4 | Understand · Apply

Learning goals

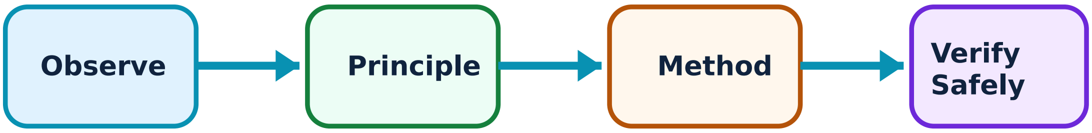
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Differential Calculus: identify the system, state its principle, follow the correct method and verify the result safely.

1. Limits−

A limit describes the value approached by a function as variable approaches a point. The official methods include direct substitution, factorisation and standard forms such as $(x^n-a^n)/(x-a)$ and sin x/x.

Study and application method

Try substitution first; if indeterminate, factorise or transform using the stated standard result, showing cancelled factor condition.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Try substitution first; if indeterminate, factorise or transform using the stated standard result, showing cancelled factor condition.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Cancellation is valid only away from excluded value; the limit describes approach, not necessarily function value.

2. Derivative concept and standard results−

Derivative measures instantaneous rate of change and is the slope of tangent in coordinate interpretation. Standard derivatives of constants, powers, exponential, log and trigonometric functions provide the calculation foundation.

Study and application method

Identify function type and variable, write standard derivative, then substitute constants/arguments carefully.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify function type and variable, write standard derivative, then substitute constants/arguments carefully.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□□ concept □□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□. □□□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Differentiate with respect to the specified variable; do not omit chain-rule factor for a nontrivial argument.

3. Rules and second derivatives−

Linearity, product and quotient rules extend differentiation to combined functions. A second derivative is derivative of first derivative and is used to describe change of rate.

Study and application method

Choose rule before expanding, keep factors grouped, derive first derivative clearly and then differentiate it for second derivative.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Choose rule before expanding, keep factors grouped, derive first derivative clearly and then differentiate it for second derivative.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: In quotient rule, denominator is squared and numerator order matters; use brackets.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Matrices and Determinants

Use the concept flow to revise observation, principle, method and verification.

Module 2: Trigonometry

Use the concept flow to revise observation, principle, method and verification.

Module 3: Coordinate Geometry

Use the concept flow to revise observation, principle, method and verification.

Module 4: Differential Calculus

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Practise transpose, matrix algebra, determinants, Cramer’s rule and applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Evaluate ratios/compound-angle problems and connect trigonometry to engineering examples.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Solve line and circle equations, angles and parallel/perpendicular problems.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Evaluate limits and derivatives and record calculus applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Matrices and Determinants

Explain Matrices, order and special types and state one correct method, application or precaution.−

A matrix is a rectangular arrangement of entries. Its order is rows by columns. The official types include row, column, square, diagonal, triangular, identity and zero matrices. Type and order determine which operations are meaningful.

Answer framework: Write order first, identify the requested type by checking positions of entries, and preserve brackets when presenting a matrix.

Important point: Matrix addition/subtraction requires equal order; do not add entries from incompatible matrices.

Explain Transpose and algebra of matrices and state one correct method, application or precaution.−

Transpose interchanges rows and columns. Matrix algebra includes scalar multiplication, equality, addition, subtraction and multiplication. Matrix multiplication is an ordered row-by-column operation and is generally not commutative.

Answer framework: For multiplication, check inner dimensions, compute each entry as row-dot-column and state final order.

Important point: AB need not equal BA; never reverse order without checking dimensions and result.

Explain Determinants, inverse and Cramer's rule and state one correct method, application or precaution.−

A determinant is associated with a square matrix. Minors and cofactors support adjoint construction; a 2×2 inverse exists when determinant is nonzero. Cramer's rule solves a compatible linear system through determinant ratios.

Answer framework: Define coefficient determinant, calculate systematically, check nonzero condition and substitute each determinant in correct variable ratio.

Important point: An inverse cannot be used when determinant is zero; write this condition explicitly.

Module 2 — Trigonometry

Explain Angles and trigonometric ratios and state one correct method, application or precaution.−

Angles may be acute, obtuse, right, straight, reflex, complete, positive or negative. Sine, cosine and tangent ratios relate sides of an acute right triangle, while reciprocal ratios complete the standard set.

Answer framework: Sketch the angle, state quadrant/sign and use a known ratio with Pythagorean relation when another ratio is needed.

Important point: Keep angle unit and quadrant convention clear; a reference-angle value may need a sign change.

Explain ASTC, reduction formulae and identities and state one correct method, application or precaution.−

Pythagorean relations and ASTC rule organise signs and relationships of trigonometric functions in all quadrants. Reduction formulae convert related angles to a standard reference.

Answer framework: Reduce angle to a familiar form, establish sign from quadrant and then apply identity with each algebraic step shown.

Important point: Do not copy only numerical standard-angle value before deciding the sign.

Explain Compound angles and state one correct method, application or precaution.−

Sine, cosine and tangent compound-angle formulae relate functions of A±B to functions of A and B. They support exact evaluation and engineering geometry calculations within official scope.

Answer framework: Write the selected formula first, substitute exact values and simplify in a controlled order.

Important point: The sign inside a compound-angle formula is not always the same as the sign outside a term; write brackets.

Module 3 — Coordinate Geometry

Explain Slope and equation of a straight line and state one correct method, application or precaution.−

Slope measures line inclination. The slope-point form creates a line equation from one point and a known slope. It connects algebraic equation to a plotted geometric object.

Answer framework: Find slope from points or angle, substitute a chosen point into point-slope form and simplify only after expanding carefully.

Important point: Vertical lines have undefined slope; do not force them into an ordinary finite-slope calculation.

Explain Angle, parallel and perpendicular lines and state one correct method, application or precaution.−

The angle between lines depends on their slopes. Parallel lines have equal slopes; perpendicular non-vertical lines have slopes whose product is −1.

Answer framework: Extract slopes from standard equation, select correct relation and check result against sketch.

Important point: A negative reciprocal rule needs care if either line is vertical or horizontal.

Explain Circles and their general equation and state one correct method, application or precaution.−

A circle is the locus of points at constant distance from centre. The general equation $x^2+y^2+2gx+2fy+c=0$ yields centre $(-g,-f)$ and radius from its coefficients when valid.

Answer framework: Compare coefficients term by term, calculate centre and radius, then verify by completing square or substitution where appropriate.

Important point: A negative radius-squared result has no real circle; report the interpretation.

Module 4 — Differential Calculus

Explain Limits and state one correct method, application or precaution.−

A limit describes the value approached by a function as variable approaches a point. The official methods include direct substitution, factorisation and standard forms such as $(x^n-a^n)/(x-a)$ and $\sin x/x$.

Answer framework: Try substitution first; if indeterminate, factorise or transform using the stated standard result, showing cancelled factor condition.

Important point: Cancellation is valid only away from excluded value; the limit describes approach, not necessarily function value.

Explain Derivative concept and standard results and state one correct

method, application or precaution.−

Derivative measures instantaneous rate of change and is the slope of tangent in coordinate interpretation. Standard derivatives of constants, powers, exponential, log and trigonometric functions provide the calculation foundation.

Answer framework: Identify function type and variable, write standard derivative, then substitute constants/arguments carefully.

Important point: Differentiate with respect to the specified variable; do not omit chain-rule factor for a nontrivial argument.

Explain Rules and second derivatives and state one correct method, application or precaution.−

Linearity, product and quotient rules extend differentiation to combined functions. A second derivative is derivative of first derivative and is used to describe change of rate.

Answer framework: Choose rule before expanding, keep factors grouped, derive first derivative clearly and then differentiate it for second derivative.

Important point: In quotient rule, denominator is squared and numerator order matters; use brackets.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Matrices and Determinants.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Trigonometry.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Coordinate Geometry.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Differential Calculus.

8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Matrix concept and types; transpose; algebra; determinants; minors/cofactors; adjoint/inverse of 2×2 matrix; Cramer’s rule..
2. Develop a complete answer or practical plan for: Angles; ratios; standard angles; Pythagorean relations; ASTC and reduction; compound angles..
3. Develop a complete answer or practical plan for: Slope and slope-point line; angle between lines; parallel/perpendicular lines; circle equation, centre and radius..
4. Develop a complete answer or practical plan for: Limits by substitution/factorisation; special forms; derivative concept and standard results; rules and second derivatives..

LAST-MILE REVISION

Quick Revision

Module 1: Matrices and Determinants

Matrix concept and types; transpose; algebra; determinants; minors/cofactors; adjoint/inverse of 2×2 matrix; Cramer’s rule.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Trigonometry

Angles; ratios; standard angles; Pythagorean relations; ASTC and reduction; compound angles.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Coordinate Geometry

Slope and slope-point line; angle between lines; parallel/perpendicular lines; circle equation, centre and radius.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Differential Calculus

Limits by substitution/factorisation; special forms; derivative concept and standard results; rules and second derivatives.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Matrices, order and special types	A matrix is a rectangular arrangement of entries.
Transpose and algebra of matrices	Transpose interchanges rows and columns.
Determinants, inverse and Cramer’s rule	A determinant is associated with a square matrix.
Angles and trigonometric ratios	Angles may be acute, obtuse, right, straight, reflex, complete, positive or negative.
ASTC, reduction formulae and identities	Pythagorean relations and ASTC rule organise signs and relationships of trigonometric functions in all quadrants.
Compound angles	Sine, cosine and tangent compound-angle formulae relate functions of $A\pm B$ to functions of A and B.
Slope and equation of a straight line	Slope measures line inclination.
Angle, parallel and perpendicular lines	The angle between lines depends on their slopes.
Circles and their general equation	A circle is the locus of points at constant distance from centre.
Limits	A limit describes the value approached by a function as variable approaches a point.
Derivative concept and standard results	Derivative measures instantaneous rate of change and is the slope of tangent in coordinate interpretation.
Rules and second derivatives	Linearity, product and quotient rules extend differentiation to combined functions.

LEARNING RESOURCES

References

Officially listed print resources

1. Schaum’s Outline of Linear Algebra.
2. Schaum’s Outline of Trigonometry.
3. Schaum’s Outline of Calculus.
4. N. P. Bali and Manish Goyal, A Textbook of Engineering Mathematics.

Officially listed web resources

Links are reproduced because they appear in the supplied syllabus. Use institutional safety and access guidance when opening external resources.